

Zhi Hong

[Email](#) | [Google Scholar](#) | [GitHub](#)

Education

South China University of Technology (SCUT)

Guangzhou, China

Dual-Degree program: B.Eng. in Automation (Control Science and Engineering)

Sep 2022 – Jul 2026

& B.Sc. in Mathematics and Applied Mathematics

GPA: 3.84/4.0 **Rank:** 3/24

Relevant Coursework: Mathematical Analysis, Advanced Algebra, Probability Theory, Mathematical Statistics, Stochastic Processes, Principles of Automatic Control, Machine Learning, Deep Learning, Reinforcement Learning

Research Interests

Large Language Models: Multi-Agent Systems, Harness Engineering, Embodied Agents

Machine Learning: Bandits, Reinforcement Learning, Federated Learning

Publications

1. **Zhi Hong***, Qian Zhang*, Jiahang Sun, Zhiwei Shang, Mingze Kong, Xiangyi Wang, Yao Shu[†], Zhongxiang Dai[†]. MASPOB: Bandit-Based Prompt Optimization for Multi-Agent Systems with Graph Neural Networks. *ICML 2026* (**Spotlight, Top 2.2% of 23,918 submissions**)

Preprints & Under Review

1. **Zhi Hong**, Qianlong Wang, Hongchen Wei, Aidan Wu, Daohong Ying, Mingze Kong, Chong Luo, Bei Liu. HARNESSFLOW: MCTS-Guided Harness Optimization for LLM Agents. Submitted to *ICLR 2027*
2. **Zhi Hong**, Jie Mao, Qianlong Wang, Daohong Ying, Shuang Qiu, Bei Liu, Zhongxiang Dai. Which LLM for Which Agent? Topology-Aware Bandit-based Model Selection for Multi-Agent Systems. Submitted to *ICLR 2027*
3. Hongchen Wei*, Daohong Ying*, **Zhi Hong***, Aidan Wu*, Bei Liu, Ruichun Ma, Yifan Yang, Qi Dai, Kai Qiu, Rui Xu, Dongdong Chen, Zhenzhong Chen, Chong Luo. OfficeTown: Generative Organizations for Evolving Agent Intelligence. Submitted to *ICLR 2027*
4. Bo Xue, **Zhi Hong**, Jiayi Li, Yuanyu Wan, Ji Cheng, Shuang Qiu. Cost-Aware Multi-Objective Bandits: Theory and Application to Budgeted LLM Configuration Evaluation. arXiv:2608.04333
5. Pingchen Lu*, **Zhi Hong***, Zhiwei Shang, Zhiyong Wang, Yikun Ban, Yao Shu, Min Zhang, Shuang Qiu, Zhongxiang Dai[†]. FedPOB: Sample-Efficient Federated Prompt Optimization via Bandits. arXiv:2509.24701
6. Jixuan Chen, Jianzhi Shen, Haoqiang Kang, **Zhi Hong**, Qingyi Jiang, Soham Bose, Yiming Zhang, Leon Leng, Amit Vyas, Lingjun Mao, Siru Ouyang, Kun Zhou, Lianhui Qin. AgentSpec: Understanding Embodied Agent Scaffolds Through Controlled Composition. arXiv:2606.14674
7. Mingze Kong, Zikun Qu, Zhongquan Zhou, Pengyu Liang, Xiang Li, Zhiwei Shang, **Zhi Hong**, Kaiyu Huang, Zhiyong Wang, Zhongxiang Dai[†]. Workflow-R1: Group Sub-sequence Policy Optimization for Multi-turn Workflow Construction. arXiv:2602.01202
8. Xiangyi Wang, Pingchen Lu, Jie Mao, Mingze Kong, **Zhi Hong**, Zhiyong Wang, Zhongxiang Dai. Linear and Neural Dueling Bandits with Delayed Feedback. arXiv:2605.26554
9. Qianlong Wang, Zhikang Xie, Xichen Ye, **Zhi Hong**, Wenzhe Cao, Yifan Wu, Cheng Jin, Weizhong Zhang, Zengfeng Huang. Entropy Forcing: Entropy-Routed Subset Selection for KV Cache in Streaming Video Generation. Submitted to *ICLR 2027*

Research Experience

Research Intern (Star Bridge Program)

Jul 2026 – Present

Advisor: Bei Liu, Microsoft Research Asia (MSRA), Visual Computing Group

Conducted research on LLM agents, including social simulation and harness engineering

Research Intern

Jan 2026 – Jul 2026

Advisor: Prof. Lianhui Qin, Computer Science and Engineering, UC San Diego

Conducted research on modular architectures to enhance capabilities of embodied LLM-based agents
Developed a specification framework for composing reusable policy components in embodied agents

Research Assistant

Sep 2024 – Jul 2026

Advisor: Prof. Zhongxiang Dai, School of Data Science, CUHK-Shenzhen

Conducted research on multi-agent systems to enhance intelligence in large language models

Honors & Awards

Outstanding Graduate – South China University of Technology	2026
First Prize – China Mathematics Competition (CMC, Category A, Province Level)	2024
National Encouragement Scholarship – Ministry of Education, China	2023 – 2025
Merit Student – South China University of Technology	2023 – 2025

Skills & Interests

Languages: English (TOEFL MyBest 101), Chinese (Native)	Programming: Python, LaTeX, C++, MATLAB
Research Skills: Reinforcement Learning, Deep Learning, Federated Learning, Bandit Algorithms	
Interests: Traveling, Basketball, Swimming	

Academic Service

Reviewer: ICLR (2027)
